

No. of Printed Pages : 4

MPC-006

M. A. (PSYCHOLOGY)
(MAPC)

Term-End Examination

June, 2023

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 Hours

Maximum Marks : 50

Note : (i) *All Sections are compulsory.*

(ii) *Use of simple calculator is permitted.*

Section—A

Note : *Answer any **two** of the following questions in about **450** words each (wherever applicable).*

2×10=20

1. Elucidate parametric and non-parametric statistics. Describe the assumptions of parametric statistics. 10
2. Explain point biserial correlation and phi-coefficient with suitable examples. 5+5

P. T. O.

3. Compute one way ANOVA for the following data :

10

Group A	Group B	Group C
4	6	10
6	7	11
7	10	12
9	12	13
2	13	14
4	14	15
7	15	17
8	17	10
9	18	2
2	3	4

Critical value for $F = 3.35$ at 0.05 level of significance.

Critical value for $F = 5.49$ at 0.01 level of significance.

4. Compute Spearman's rho for the following data :

10

Data A	Data B
16	15
19	17
18	16
10	9
12	10
13	12
17	19
9	8
7	6
5	4

Section—B

Note : Answer any **four** question in about **250** words each (wherever applicable).

4×6=24

5. Explain the characteristics of normal probability curve. 6
6. Compute Chi-square for the following data : 6

Responses

Gender	Yes	No
Male	15	5
Female	10	10

7. Discuss the importance and application of standard error of means. 6
8. Compute Mann-Whitney U test for the following data : 6

Group A	Group B
21	12
30	16
26	17
32	19
41	20
35	24
15	31

9. Explain the concept of interactional effect. Discuss merits and demerits of two way ANOVA. 2+6

Section—C

Note : Answer any **two** of the following in about 100 words each. $2 \times 3 = 6$

10. Hypothesis testing. 3
11. Outliers and curvilinearity. 3
12. Wilcoxon matched pair signed rank test. 3